

RELATIONS - HOMEWORK

1. Write down the ordered pairs belonging to each of the following relations on $\mathbb{N}$:

(a) $R = \{(x, y) : 2x + y = 9\}$

(b) $S = \{(x, y) : x + y < 7\}$

2. Write down the ordered pairs belonging to the following binary relations between the sets $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5\}$.

(a) $U = \{(x, y) : x^2 \leq y\}$

(b) $V = \{(x, y) : x + y \text{ is odd}\}$

(c) $W = \{(x, y) : y - x < 3\}$

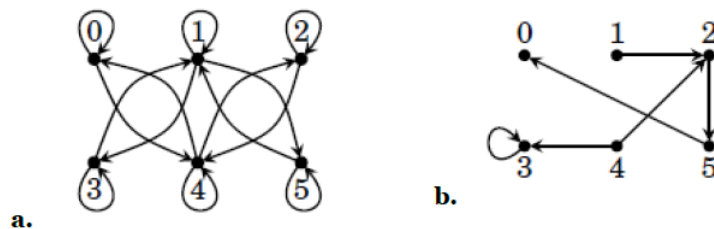
3. Draw the directed graph representation and write down the matrix representation of the relations U , V and W in Problem 2.

4. Let R be a relation between the sets $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$ that has the following matrix representation.

$$\begin{bmatrix} T & F & T & F \\ T & F & T & F \\ F & T & T & F \end{bmatrix}$$

List the set of ordered pairs and draw the digraph representation of R .

5. The following are digraphs on relations R on A . List the elements of R and A in each case.



6. Let R be a relation on $\{1, 2, 3, 4\}$ given by uRv if and only if $u + 2v$ is odd. Represent R in each of the following ways:

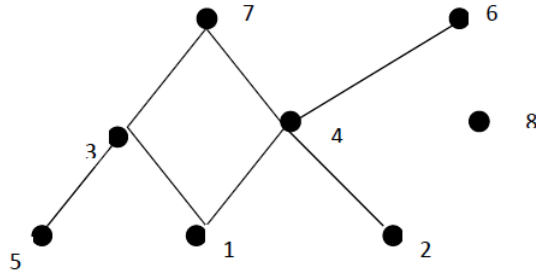
(a) as a set of ordered pairs

(b) in graphical form (digraph)

(c) in matrix form

7. Let R be a relation on $\{1, 2, 3, 4, 5\}$ given by xRy if and only if $2x \geq 3y$. Represent R in each of the following ways:
- (a) as a set of ordered pairs
 - (b) in graphical form (digraph)
 - (c) in matrix form
8. Determine if each of the following relations on the set of people is reflexive, symmetric, antisymmetric or transitive. Hence, determine if each of the relations is an equivalence relation, a partial order or neither:
- (a) x has the same parents as y
 - (b) x is a brother of y
 - (c) x is older or younger than y
 - (d) x is at least as clever as y
9. Determine if each of the following relations on $\mathbb{Z}$ is reflexive, symmetric, antisymmetric or transitive. Hence, determine if each of the relations is an equivalence relation, a partial order or neither:
- (a) $x + y$ is an odd integer
 - (b) $x + y$ is an even integer
 - (c) xy is an odd integer
 - (d) $x + xy$ is an even integer
 - (e) $xy > 0$
10. For each of the following relations on the set $\{x : x \in \mathbb{Z} \text{ and } 1 \leq x \leq 12\}$ list the ordered pairs belonging to the relation:
- (a) $R = \{(x, y) : xy = 9\}$
 - (b) $S = \{(x, y) : 2x = 3y\}$
 - (c) the transitive closure of R
 - (d) the transitive closure of S
11. Determine each of the reflexive, symmetric and transitive closures of the relation on the set $\{a, b, c, d\}$.
- $$\{(a, a), (b, b), (c, c), (a, c), (a, d), (b, d), (c, a), (d, a)\}$$
12. Let $R = \{(x, y) : x \text{ is a divisor of } y\}$ be a partial order on the set $A = \{1, 2, 3, 6, 10, 15, 30\}$. Draw a table of predecessors and immediate predecessors. Hence, draw the corresponding Hasse diagram. State which are the minimal and maximal elements.

13. The following is the Hasse diagram of a relation R on A . List the elements of the relation.



14. Let R be the relation between the sets $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$ given by:

$$R = \{(1, 1), (2, 3), (2, 4), (3, 1), (3, 4)\}$$

Let S be the relation between the sets B and $C = \{1, 2\}$ given by:

$$S = \{(1, 1), (1, 2), (2, 1), (3, 1), (4, 2)\}$$

Calculate R^{-1} , S^{-1} , $S \circ R$, $R \circ S$, $R^{-1} \circ S^{-1}$ and $(S \circ R)^{-1}$ and draw the corresponding digraphs.

15. Let R denote the relation *is a parent of* and S denote the relation *is a brother of* on the set of all people. Give concise verbal descriptions of the relations R^{-1} , S^{-1} , $R \circ S$, $S^{-1} \circ R^{-1}$ and $R \circ R$.